ATS

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Chemistry Informal Diagnostics (ID)

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Stream: Advanced

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Subtopic 6.2 – Strength of Acids and Bases

	Subtopic 6.2 – Strength of Acids and Bases
	Inspire Chemistry – Module 17 – Lesson 2: Strength of Acids & Bases

KPI 6.2.2 – Define the acid ionization constant, Ka, while writing the ionization constant expression for different weak acids.

DOK 1: Recall & Reproduction

1. Define the acid ionization constant, Ka, and write the ionization constant expression for acetic acid, CH3COOH.
Answer:
The acid ionization constant, Ka, is a measure of the strength of a weak acid in solution, representing the equilibrium constant for its dissociation in water.

For acetic acid:

CH3COOH(aq)⇌CH3COO−(aq)+H+(aq)

The expression for Ka is:

Ka=[CH3COO−][H+]/[CH3COOH]

KPI 6.2.1 – Compare strong and weak acids (using examples, particulate diagrams, and ionization equations).

DOK 2: Skills & Concepts

2. Compare strong and weak acids in terms of their ionization in water. Provide an example of each with its corresponding ionization equation.
Answer:

- **Strong acids** completely ionize in water. Example:

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

- **Weak acids** only partially ionize in water. Example:

$HF(aq) \rightleftharpoons H^+(aq) + F^-(aq)$

Key Difference: A strong acid exists entirely as ions in solution, whereas a weak acid exists in dynamic equilibrium with its undissociated molecules.

KPI 6.2.3 – Relate the strength of weak acids to the numerical values of Ka.

DOK 3: Strategic Thinking

3. Relate the strength of a weak acid to the numerical value of Ka. Support your claim with evidence from data.
Answer:

- The **larger** the Ka value, the **stronger** the weak acid because more of it ionizes.
- The **smaller** the Ka, the **weaker** the acid, meaning it ionizes less.

For example:

- **HF ($Ka=6.6\times10^{-4}$)** is a stronger weak acid than **HCN ($Ka=6.2\times10^{-10}$)** because HF ionizes more.
- **Evidence:** A high Ka corresponds to a lower pKap (pKa=−logKap), which means greater acidity.

KPI 6.2.3 – Relate the strength of weak acids to the numerical values of Ka.

DOK 2: Skills & Concepts

4. Multiple Choice: Which of the following acids is the weakest based on their Ka values?
(A) **Formic acid, $Ka=1.8\times10^{-4}$.**
(B) **Nitrous acid, $Ka=4.6\times10^{-4}$.**
(C) **Hydrocyanic acid, $Ka=6.2\times10^{-10}$.** ✓
(D) **Acetic acid, $Ka=1.8\times10^{-5}$.**

Explanation: The weakest acid has the smallest Ka, meaning it ionizes the least in water.

KPI 6.2.5 – Explore and explain the relationship between the strength of an acid and its conjugate base and the strength of a base and its conjugate acid.

5. Explain how the strength of an acid is related to the strength of its conjugate base. Use data to justify your response.
Answer:

- A strong acid has a very **weak conjugate base** because it readily donates H+H^+.
- A weak acid has a **stronger** conjugate base because it holds onto H+H^+ more tightly.

For example:

- **HCl (strong acid) → Cl^- (very weak base)**
- **HF (weak acid) → F^- (relatively strong base)**

Since **$Ka \cdot Kb = Kw (1.0 \times 10{-}14)$** a higher KaK_a (strong acid) means a lower Kb(weak conjugate base), and vice versa.

KPI 6.2.6 – Investigate the base ionization constant, Kb, while writing the ionization constant expression for different weak bases.

6. Define the base ionization constant, Kb, and write the ionization constant expression for ammonia, NH_3.
Answer:
The base ionization constant, Kb, measures the strength of a base in solution. It represents the equilibrium constant for the reaction of a weak base with water.

For ammonia:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

The expression for Kb is:

$$Kb = [NH_4^+][OH^-] / [NH_3]$$

KPI 6.2.4 – Compare strong and weak bases (using examples, particulate diagrams, and ionization equations).

7. Compare the ionization of a strong base (NaOH) and a weak base (NH_3). Provide particulate diagrams to illustrate the difference.
Answer:

- **NaOH (strong base)** fully dissociates into Na^+ and OH^- in water.
- **NH_3 (weak base)** only partially reacts with water to form NH_4^+ and OH^-, leaving many NH_3 molecules unreacted.

KPI 6.2.7 – Predict the strength of weak bases based on their KbK_b values.

8. Predict the strength of weak bases based on their KbK_b values. Justify your answer with an example.
Answer:

- **Larger Kb → Stronger weak base**
- **Smaller Kb → Weaker weak base**

For example:

- **Ammonia, NH_3, $Kb = 1.8 \times 10^{-5}$** is a **stronger weak base** than
- **Aniline, $C_6H_5NH_2$, $Kb = 3.8 \times 10^{-10}$** because it ionizes more.

1. Which of the following is a weak acid?

√ A. H_2CO_3

B. NaOH

C. H_2SO_4

D. NH_3

2. Which of the following is/are **true** about nitric acid?

I. A binary acid
II. A strong electrolyte
III. Ionizes completely in water

A. II only

B. III only

C. I and II only

√ D. II and III only

3. Which of the following equations represents the ionization of hydrochloric acid?

√ A. $HCl + H_2O \rightarrow H_3O^+ + Cl^-$

B. $HCl + H_2O \rightleftarrows H_3O^+ + Cl^-$

C. $HClO_4 + H_2O \rightarrow H_3O^+ + ClO_4^-$

D. $HClO_4 + H_2O \rightleftarrows H_3O^+ + ClO_4^-$

4. The acid ionization constants, Ka, of three different acids **HX**, **HY** and **HZ** are given below. Which of the following correctly lists the three acids in **decreasing** order of their acid strength?

HX	$Ka = 1.6 \times 10^{-5}$
HY	$Ka = 2.4 \times 10^{-3}$
HZ	$Ka = 3.0 \times 10^{-7}$

A. $HX > HY > HZ$

√ B. $HY > HX > HZ$

C. $HZ > HY > HX$

D. $HX > HZ > HY$

5. The correct ionization constant expression of NH_3 is ______.

√ A. $K_b = \frac{[NH_4^+]\,.\,[OH^-]}{[NH_3]}$

B. $K_b = \frac{[NH_4]^+.\,[H^+]}{[NH_3]}$

C. $K_b = \frac{[NH_3]}{[NH_4^+]\,.\,[OH^-]}$

D. $K_b = \frac{[NH_3]}{[NH_4^+]\,.\,[H^+]}$

6. Consider the Ka values for acids X and Y given below.

Acid	**X**	**Y**
Ka at 298 K	6.2×10^{-10}	6.3×10^{-4}

Acid **X** is __________ than acid **Y**, because it has a __________ value of *K*a.

√ A. weaker smaller

B. stronger smaller

C. weaker larger

D. stronger larger

7. **HX** and **H_2Y** are **Arrhenius** acids. Use the information below to answer questions **a** and **b**.

HX	**Strong acid**
H_2Y	**Weak acid**

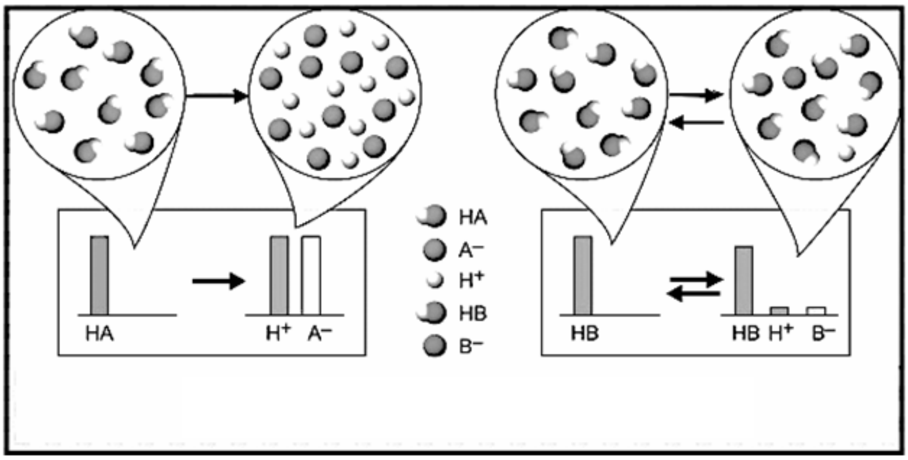
a) Write the **ionization equation** for acids **HX** and **H_2Y**.

Acid	Ionization equation
HX	$HX + H_2O \rightarrow H_3O^{+} + X^{-}$
H_2Y	$H_2Y + H_2O \rightleftharpoons H_3O^{+} + HY^{-}$

b) Which acid **HX** or **H_2Y** produces **higher concentration of hydrogen ions** when dissolved in water? Explain your answer.

HX. It completely dissociates or ionizes in water

8. Use the following diagram that represent two acids **HA** and **HB** to answer questions **a** and **b**.



a) Write the ionization equations for **HA** and **HB**.

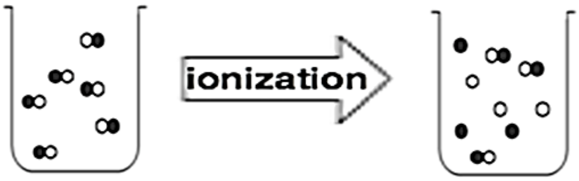
$HA \rightarrow H^{+} + A^{-}$
$HB \rightleftharpoons H^{+} + B^{-}$

b) Which is a strong acid: HA or HB? Justify your answer.

HA
It is completely ionized or dissociated in water

9. The figure below shows the particulate diagrams of the ionization process of three different acids: **HX**, **HY** and **HZ**. Answer questions **a** - **c**.

Acid HX



Acid HY

Acid HZ

a) Identify the **strongest** acid? Justify your answer.

Strongest acid is HX. It ionizes/dissociates completety in aqueous solution.

b) Identify the **weakest** acid? Justify your answer.

HZ is the weakest acid. It ionizies/dissociates the least.

c) Arrange the acids in **increasing** order of their strength. Explain your answer.

HZ < HY < HX

10. Based on your knowledge about acids and bases, classify each of the following substances as a strong or a weak acid, strong or a weak base and write its ionization equation.

Substance	Strong acid, weak acid , strong base or weak base	Ionization equation
CH_3COOH	Weak acid	$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$
NH_3	Weak base	$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

	Subtopic 6.3 – Hydrogen Ions & pH
	Inspire Chemistry – Module 17 – Lesson 3: Hydrogen Ions & pH

6.3.1 Define the ion-product constant for water, Kw, while writing its expression and value at 25°C.

Question: What is the ion-product constant for water, K_w, at 25°C?

Answer: The ion-product constant for water, K_w, is the equilibrium constant for the self-ionization of water:

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

At 25°C, the value of K_w is 1.0×10^{-14}.

✅ **Scientific Explanation:** Since water is amphiprotic, a small fraction of water molecules dissociate into hydrogen ions and hydroxide ions. The equilibrium constant expression is:

$$K_w = [H^+][OH^-]$$

Since pure water contains equal concentrations of H^+ and OH^-, we deduce:

$$[H^+] = [OH^-] = \sqrt{K_w} = 1.0 \times 10^{-7}\ \mathrm{M}$$

6.3.2 Predict the hydronium and hydroxide ion concentrations at a given temperature and vice versa using KwK_w.

Question: If the concentration of hydronium ions in a solution is 2.0×10^{-5} M at 25°C, calculate the concentration of hydroxide ions.

Answer:

$$K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$$

Solving for $[OH^-]$:

$$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1.0 \times 10^{-14}}{2.0 \times 10^{-5}} = 5.0 \times 10^{-10}\ \mathrm{M}$$

✅ **Scientific Explanation:** This calculation is based on **Le Chatelier's principle**, where increasing $[H_3O^+]$ decreases $[OH^-]$ to maintain equilibrium.

6.3.3 Define pH and write its mathematical formula.

DOK Level 1: Recall

Question: Define pH and write its mathematical formula.
Answer: pH is a measure of the **acidity** of a solution, calculated as:

$$pH = -\log[H^+]$$

✅ **Scientific Explanation:** The logarithmic scale compresses large differences in [H+] into a manageable range (0-14). The phenomenon of **self-ionization of water** ensures that **neutral** water has pH=7, while acids have lower pH values.

6.3.4 Explain what the pH scale is.

Question: What is the pH scale, and what does it measure?
Answer: The pH scale ranges from **0 to 14**, where:

- **Acidic solutions**: $pH < 7$ (higher $[H^+]$ concentration)
- **Neutral solutions**: $pH = 7$
- **Basic solutions**: $pH > 7$ (higher $[OH^-]$ concentration)
  ✅ **Scientific Explanation:** The pH scale is **logarithmic**, meaning that each pH unit represents a **10-fold** change in $[H^+]$. This explains why a **pH of 4 is 1000 times more acidic** than a pH of 7.

6.3.5 Define pOH and write its mathematical formula.

DOK Level 1: Recall

Question: What is pOH, and how is it different from pH?

Answer:

$$pOH = -\log[OH^-]$$

✅ **Scientific Explanation:** Since $K_w = 1.0 \times 10^{-14}$, the relationship between **pH and pOH** is:

$$pH + pOH = 14$$

A high pOH means a **low concentration of OH^-**, indicating an **acidic solution**.

6.3.6 Describe the relation between pH and pOH and perform calculations.

DOK Level 3: Strategic Thinking

Question: If a solution has a pH of **4.8**, what is its pOH?
Answer:

$$pOH = 14 - pH = 14 - 4.8 = 9.2$$

✅ **Scientific Explanation:** This follows the **pH-pOH relationship** derived from the ion-product constant of water.

6.3.7 Relate acidity and basicity to hydronium and hydroxide concentrations.

DOK Level 2: Conceptual Understanding

Question: If the pH of a solution is **9**, what can you conclude about its acidity or basicity?
Answer: Since **pH > 7**, the solution is **basic** with a **low [H+] concentration**.
✅ **Scientific Explanation:** The relationship between **hydronium and hydroxide ions** ensures that an increase in pH corresponds to a **decrease in acidity**.

6.3.8 Calculate the pH of a solution given [H+] or [OH−].

DOK Level 3: Problem Solving

Question: A solution has [H+]=1.0×10−4 M. What is its pH?
Answer:

$pH=-\log(1.0\times10^{-4})=4.00$

✅ **Scientific Explanation:** This demonstrates how **logarithmic calculations** apply to acid-base chemistry.

6.3.9 Predict pOH when given [H+] or [OH−]

DOK Level 3: Problem Solving

Question: If $[OH^-]=1.0\times10^{-6}$. M, what is the pOH and corresponding pH?
Answer:

pOH=8.00

✅ **Scientific Explanation:** This confirms that **low [OH−] means high acidity**.

6.3.10 Predict pH and pOH from [OH−].

DOK Level 3: Problem Solving

Question: A solution has $[OH^-] = 2.0 \times 10^{-5}$ M. Determine the pH.

Answer:

$$pOH = -\log(2.0 \times 10^{-5}) = 4.70$$

$$pH = 14 - 4.70 = 9.30$$

✅ **Scientific Explanation:** This shows the inverse relationship between $[OH^-]$ and pH.

11. What is the pOH of a solution that has a pH = 6.0 at 25°C?

A. 2.0

B. 4.0

C. 6.0

√ D. 8.0

12. Which of the following is always **true** about solutions at 25 °C?

√ A. $[H^+] \times [OH^-] = 1.0 \times 10^{-14}$

B. $[H^+] > [OH^-]$

C. $pH\ + pOH = Kw$

D. $Kw = 1.0 \times 10^{-10}$

13. As the concentration of hydrogen ions decreases, the solution becomes more ______ and the pH ______.

√ A. basic increases

B. acidic increases

C. basic decreases

D. acidic decreases

14. At 25°C, a solution has $[\mathrm{H}^+] = 3.2 \times 10^{-4}\mathrm{M}$. The pOH of this solution is _________.

A. 2.00

B. 3.50

C. 7.00

√ D. 10.50

15. As a neutral solution becomes more acidic, the $[\mathrm{H}^+]$ ________,$[\mathrm{OH}^-]$________and the pH ________.

A. decreases increases decreases

B. increases decreases increases

C. increases increases increases

√ D. increases decreases decreases

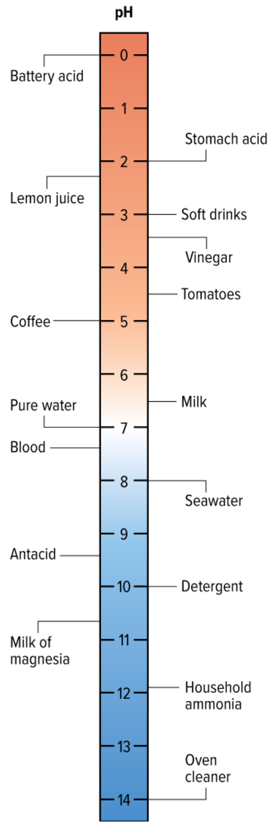
16. Based on the figure at right, which substance would have the lowest pOH?

A. battery acid

B. pure water

C. blood

√ D. oven cleaner



Use the following information to answer questions 95 – 98.
The table below shows information about five different solutions.

A.	*Solution K*	$[OH^-] = 2.0 \times 10^{-5}$
B.	*Solution L*	$pH = 3.1$
C.	*Solution M*	$[H^+] = 2.0 \times 10^{-2}$
D.	*Solution N*	$[OH^-] = 1.0 \times 10^{-7}$
E.	*Solution O*	$pOH = 3.1$

The **most acidic** solution is ________. C

The **most basic** solution is __________. E

The **neutral** solution is ________________. D

The **least basic** solution is _____________. C

17. Calculate the pH of the following solutions.

a) 1.0M HI — pH= 0.00
b) 0.050M HNO_3 — pH = 1.30
c) 1.0M KOH — pH = 14.00
d) $2.4 \times 10^{-5} M$ $Mg(OH)_2$ — pH = 9.68

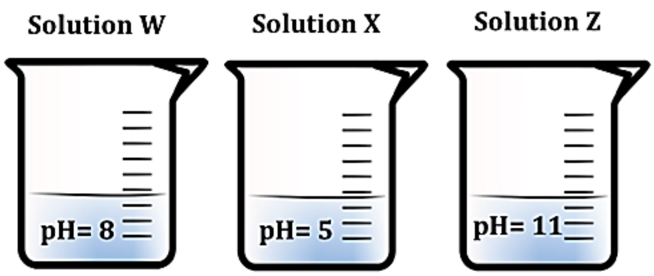
18. Determine the pH of a solution that contains $1.0 \times 10^{-9} mol$ of OH^- ions per liter.

pH = 5.00

19. The pH of a tomato is approximately 4.50. What are $[H^+]$ and $[OH^-]$ in a sample of pure tomato juice?

$[H^+] = 3.2 \times 10^{-5}$M, $[OH^-] = 3.2 \times 10^{-10}$M

20. The diagram below shows the pH of three different solutions **W**, **X**, and **Z**. Answer questions a and b.



a) Which solution is more basic **W** or **Z**?

Solution Z, it has the highest pH.

b) Based on the pH values, classify solutions **W**, **X**, and **Z** as acidic, basic, or neutral solutions.

Solution W: __________basic

Solution X: __________acidic

Solution Z: __________basic

21. Formic acid is used to process latex tapped from rubber trees into natural rubber. The pH of a 0.100M solution of formic acid (HCOOH) is 2.38. What is K_a for HCOOH?

$[H^+] = 4.2 \times 10^{-3} M$
$[HCOO^-] = [H^+] = 4.2 \times 10^{-3} M$
$[HCOOH] = 0.100 M - 4.2 \times 10^{-3} M = 0.096 M$
$K_a = \frac{[H^+][HCOO^-]}{[HCOOH]}$

$K_a = \frac{(4.2 \times 10\text{-}3)(4.2 \times 10\text{-}3)}{(0.096)} = 1.8 \times 10^{-4}$

22. As a neutral solution becomes more acidic, the $[H^+]$ ________, $[OH^-]$ ________and the pH ________.

A.	decreases	increases	decreases
B.	increases	decreases	increases
C.	increases	increases	increases
D.	decreases	decreases	increases
√ E.	increases	decreases	decreases

23. Which of the following is/are **true** about self-ionization of water?
I. At 25°C, $K_w = 1.0 \times 10^{-14}$
II. $K_w = [H^+][OH^-]$
III. $K_w = \frac{[H^+][OH^-]}{[H_2O]}$

A. I only

B. II only

C. III only

√ D. I and II only

E. I and III only

24. The pH of four different solutions (**K**, **L**, **M** and **N**) are given as shown in the table below.
Answer questions **a** - **d**.

Solution	*pH*
K	6.1
L	9.3
M	1.7
N	7.0

a) Suggest a device that can be used to measure the pH of the solutions above.

pH meter

b) Which solution has the lowest concentration of hydroxide ions?

M

c) Which solution is neutral?

N

d) Arrange the solutions above from the **least basic to the most basic**.

M < K < N < L

25. Use the information below about solutions **X**, **Y** and **Z** to answer questions **a** and **b**.

Solution X	Solution Y	Solution Z
pH = 5.60	$[H_3O^+] = 4.2 \times 10^{-3}\ M$	pOH=12.3

a) Calculate the pH of solutions **Y** and **Z.**

pH of solution Y = $-\log [H_3O^+] = -\log 4.2 \times 10^{-3} = 2.38$

pH of solution Z= $= 14 - pOH = 14 - 12.3 = 1.7$

b) Which of the solutions **X**, **Y**, and **Z** is the **most acidic**? Justify your answer.

Solution Z, it has the lowest pH or the highest $[H_3O^+]$

26. Consider a solution of barium hydroxide, **$Ba(OH)_2$**, with a concentration of **2.50×10^{-3}** ***M***. Answer questions **a – c**.

a) Calculate the concentration of hydronium ions $[H_3O^+]$.

$Ba(OH)_2 \rightarrow Ba^{2+} + 2\,OH^-$
1 : 1 : 2

$[OH^-] = 2 \times 2.50 \times 10^{-3} = 5.00 \times 10^{-3}\ M$

$[H^+] = \frac{K_w}{[OH^-]} = \frac{1.00 \times 10^{-14}}{5.00 \times 10^{-3}} = 2.00 \times 10^{-12}\ M$

b) Calculate the pH of $Ba(OH)_2$ solution.

$pH = -\text{Log}\,[H_3O^+]$
$pH = -\log(2.00 \times 10^{-12}) = 11.699$

c) Indicate whether barium hydroxide solution is **acidic**, **basic**, or **neutral**. Justify your answer.

Basic solution, because $[OH^-] > [H_3O^+]$

Subtopic 6.4 – Neutralization KPIs 6.4.1-6.4.6

	Subtopic 6.4 – Neutralization
	Inspire Chemistry – Module 17 – Lesson 4: Neutralization

6.4.1 Define neutralization reaction while writing the neutralization equation (Complete ionic and net ionic equations).
DOK Level 1: Recall

Question: What is a neutralization reaction? Write its complete and net ionic equations.

Answer: A **neutralization reaction** is a reaction between an **acid and a base**, producing **water and a salt**.

✅ **General Equation:**

$$\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$$

Example: Hydrochloric acid (HCl) reacts with sodium hydroxide ($NaOH$):

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

✅ **Complete Ionic Equation:**

$$H^+(aq) + Cl^-(aq) + Na^+(aq) + OH^-(aq) \rightarrow Na^+(aq) + Cl^-(aq) + H_2O(l)$$

✅ **Net Ionic Equation:**

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

📌 **Scientific Explanation:** This reaction illustrates the **self-ionization of water**, where hydrogen ions and hydroxide ions **neutralize** to form water molecules.

6.4.2 Define titration and titrant.
DOK Level 1: Recall

Question: What is titration? What is a titrant?

Answer:

- **Titration** is a laboratory technique where a solution of known concentration (titrant) is added to a solution of unknown concentration **to determine its molarity**.
- **Titrant** is the solution **of known concentration** added during titration.

📌 **Scientific Explanation:** Titration follows **stoichiometry**, as the moles of acid and base react in a **fixed ratio**.

6.4.3 Carry out an acid-base titration through an experiment to identify the endpoint.
DOK Level 3: Strategic Thinking

Experiment: Titration of HCl with NaOH

Materials:

- **Burette** (filled with **0.1 M NaOH**)
- **Erlenmeyer flask** (containing **HCl of unknown concentration** + **phenolphthalein indicator**)
- **Pipette** and **pH meter**

Procedure:

1. **Fill the burette** with the NaOH solution.
2. **Record the initial volume** of NaOH.
3. **Slowly add NaOH** to the HCl while stirring.
4. **Observe the color change** from **colorless to light pink**, indicating the endpoint.
5. **Record the final volume** and use the formula:

$$M1V1=M2V2$$

to determine the concentration of HCl.

📌 **Scientific Explanation:**
At the **endpoint**, moles of acid = moles of base. The indicator changes color due to **pH shift** near neutralization.

6.4.4 Explain the difference between the equivalence point and the endpoint of the titration process.
DOK Level 2: Conceptual Understanding

Question: What is the difference between the **equivalence point** and the **endpoint** in titration?

Answer:

- **Equivalence Point**: The **exact point** where the moles of **acid and base are equal**. Determined using a **pH meter**.
- **Endpoint**: The **visible color change** of an indicator, which occurs **slightly after the equivalence point**.

📌 **Scientific Explanation:**
In a **strong acid-strong base titration**, the equivalence point occurs at **pH 7**, while the **endpoint depends on the indicator**.

6.4.5 Describe the titration curve of a strong acid with a strong base.

DOK Level 3: Strategic Thinking

Question: Describe the titration curve for **HCl vs. NaOH**, including the type of salt, pH at the equivalence point, indicator used, and volume of titrant needed.

Answer:
✅ **Titration Curve Features:**

1. **Before Equivalence Point**: Excess **HCl**, pH is **low** (~2).
2. **Equivalence Point**: pH=7pH = 7, forming **neutral NaCl** salt.
3. **After Equivalence Point**: Excess **NaOH**, pH **rises sharply**.

✅ **Indicator Used: Phenolphthalein** (colorless in acid, pink in base).

✅ **Volume of NaOH required:** Found using:

$$M1V1=M2V2$$

📌 **Scientific Explanation:**
A **strong acid and strong base** fully dissociate, so **the salt does not hydrolyze**, keeping the equivalence pH at **7**.

6.4.6 Investigate the titration curve of a weak acid with a strong base.

DOK Level 3: Strategic Thinking

Question: How does the titration curve for **acetic acid (CH_3COOH) with NaOH** differ from a strong acid-base titration?

Answer:
✅ **Titration Curve Features:**

1. **Before Equivalence Point**: The solution is **buffered**, with **pH > 2** due to partial ionization of CH_3COOH.
2. **Equivalence Point**:
   - **pH ≠ 7**, but **pH ≈ 8.7** due to **hydrolysis of acetate ion (CH_3COO^-)**.
   - The **salt formed** (sodium acetate) is **basic**.
3. **After Equivalence Point**: Excess NaOH leads to **rapid pH increase**.

✅ **Indicator Used: Phenolphthalein** (pH transition: 8.2 – 10.0).

✅ **Volume of NaOH required:**
Same as **strong acid-strong base** titration.

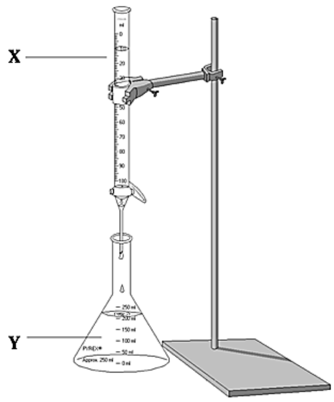
📌 **Scientific Explanation:**
Since CH_3COOH is a **weak acid**, it does **not fully ionize**. The conjugate base, CH_3COO^-, undergoes hydrolysis, shifting the **equivalence point above 7**.

Summary Table of Titration Differences

Titration Type	Type of Salt Formed	Equivalence Point pH	Indicator Used	Color Change
Strong Acid + Strong Base	Neutral salt (e.g., NaCl)	**7.0**	**Phenolphthalein or Methyl Red**	Colorless → Pink
Weak Acid + Strong Base	Basic salt (e.g., CH_3COONa)	**>7 (around 8.7)**	**Phenolphthalein**	Colorless → Pink

27. A 20.0 mL of ethanoic acid are transferred to a piece of apparatus, **Y**, where a few drops of an appropriate indicator are added. The solution is then titrated with a 0.500 mol/L solution of sodium hydroxide, NaOH. The end point is observed when 20.50 mL of NaOH is added to the solution. Answer questions

28. Which of the following correctly lists the names of pieces of apparatus **X** and **Y**?



		X	Y
	A.	Beaker	Burette
	B.	Conical flask	Burette
	C.	Burette	Pipette
	D.	Conical flask	Burette
√	E.	Burette	Conical flask

29. Which of the following indicators can be used in this titration?

I. Phenol red
II. Methyl orange
III. Bromothymol blue

√ A. I only
B. II only
C. III only
D. I and II only
E. I and III only

30. Which of the following could be the value of the pH at the equivalence point?

A. 0
B. 4
C. 7
√ D. 9
E. 12

31. What is the molar concentration in mol/L of CH_3COOH solution?

A. 0.125

B. 0.250

C. 0.488

D. 0.500

√ E. 0.513

32. What is the molarity of a nitric acid solution if 43.33 mL of 0.1000 *M* KOH solution is needed to neutralize 20.00 mL of the acid solution?

Number of moles of acid = number of moles of base
$C_aV_a = C_bV_b$
$C_a \times (20.00/1000) = 0.1000 \times (43.33/1000)$
$C_a = 0.2167$ *M*

33. What is the concentration of a household ammonia cleaning solution if 49.90 mL of 0.5900 *M* HCl is required to neutralize 25.00 mL of the solution?

Number of moles of acid = number of moles of base
$C_aV_a = C_bV_b$
$0.5900 \times (49.90/1000) = C_b \times (25.00/1000)$
$C_b = 1.178$ *M*

34. Explain the difference between equivalence point and end point of a titration.

Equivalence point is the point at which moles of hydrogen ions from the acid equal the moles of hydroxide ions from the base

End point is the point at which the indicator that is used in titration changes color

35. When might a pH meter be better than an indicator to determine the end point of an acid-base titration?

A pH meter could be used when there is no acid-base indicator that changes color at or near the equivalence point, or when such an indicator is not available.

36. A 25.00 mL of HCl solution, is titrated with 0.150 mol/L of sodium hydroxide solution, NaOH, solution. Answer questions **a – d.**

a) Which substance is the titrant in this experiment?

NaOH

b) If the concentration of HCl is 0.0600 mol/L, what is the volume of NaOH needed at the equivalence point?

At the equivalent point, number of moles of acid=number of moles of base
$C_aV_a = C_bV_b$
$0.0600\ (25) = 0.150\ (V)$
$V_{NaOH} = 10.0$ mL

c) What type of salt is formed during this titration?

Neutral

37. In an acid-base titration, 45.78 mL of a sulfuric acid solution is titrated to the end point by 74.30 mL of 0.4338 *M* sodium hydroxide solution. What is the molarity of the sulfuric acid solution?

$H_2SO_{4\,(aq)} + 2\ NaOH_{(aq)} \rightarrow Na_2SO_{4\,(aq)} + 2H_2O_{(l)}$
1 : 2
n_a : n_b

$2\ n_a = n_b$
$2 \times C_a \times V_a = C_b \times V_b$
$2 \times C_a \times (45.78/1000) = 0.4338 \times (74.30/1000)$
$C_a = 0.352\ M$

38. A student performed a titration experiment where he titrated 25.0 mL hydrochloric acid solution, HCl, with 1.0 *M* sodium hydroxide solution, NaOH. The amount of sodium hydroxide needed to reach the equivalence point was 15.0 mL. Answer questions **a** – **d**.

a) Which solution is the titrant: hydrochloric acid or sodium hydroxide?

Sodium hydroxide

b) Identify the type of salt formed. Justify your answer.

Neutral
It results from the reaction between a strong acid and a strong base

c) Calculate the concertation of hydrochloric acid at the equivalence point.

At the equivalent point, number of moles of acid=number of moles of base
$C_aV_a=C_bV_b$
C_a (25.0 / 1000) =1.0 (15.0/1000)
C_a =0.600 *M*

d) Write the name of an indicator that can be used in this experiment.

Phenolphthalein

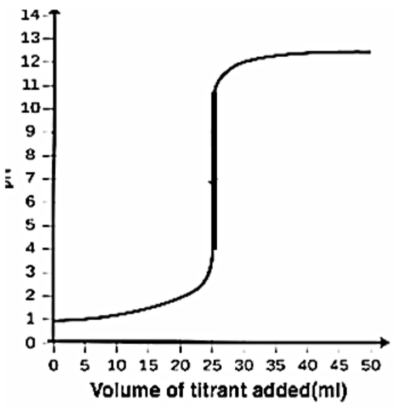
39. A student performed an experiment to determine the concentration of sulfuric acid solution, H_2SO_4. He titrated the acid with a standardized solution of potassium hydroxide, KOH. Answer questions **a** and **b**.

a) **Name** the type of reaction occurring between sulfuric acid and potassium hydroxide.

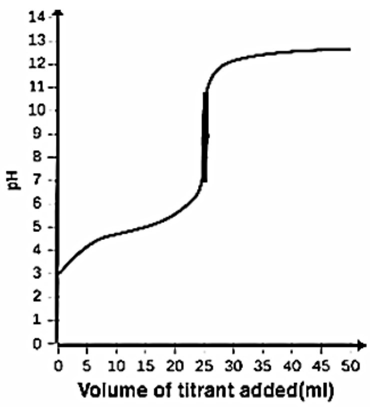
Neutralization reaction

40. A student performed three different titration experiments (**W, X** and **Y**) where he used three different acids as shown below. The student used sodium hydroxide solution, NaOH, in the three experiments. Answer questions **a – c**.

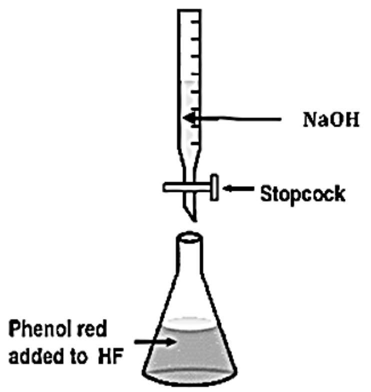
Experiment W
Acid: HCl



Experiment X
Acid: CH_3COOH



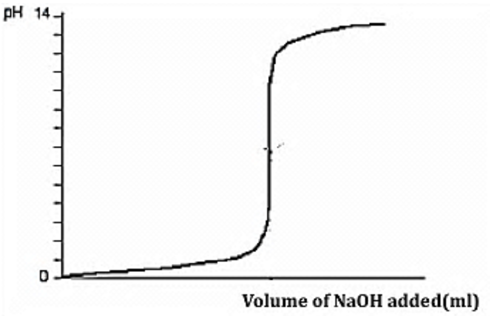
Experiment Y
Acid: HF



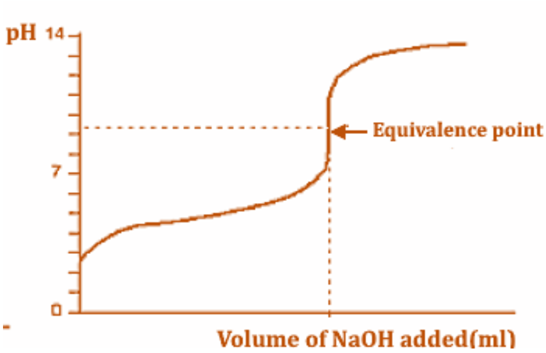
a) **Name** the titrant used in Experiment **Y.**

Sodium hydroxide

b) For Experiment **Y**, the student plotted the graph below. Did the student plot the graph correctly? If not, sketch the titration graph for Experiment **Y**.



No



1. In a neutralization reaction, an acid and a base react to produce a(n)________and water.

salt

41. Explain the difference between the equivalence point and the end point of a titration.

The equivalence point is the pH at which the moles of H+ ions from the acid equal the moles of OH− ions from the base. The endpoint is the point at which the indicator used in a titration changes color.

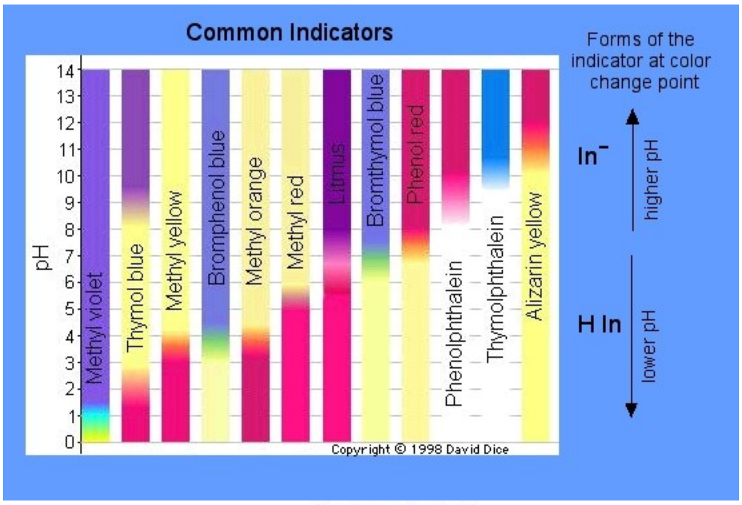
42. What happens when an acid is added to a solution containing the HF/F− -buffer system?

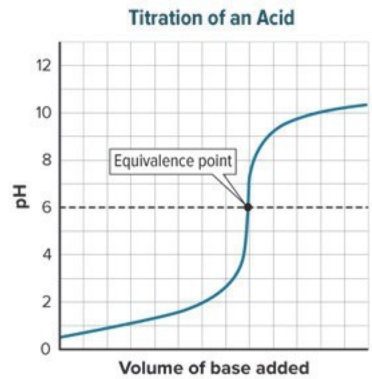
The acid produces hydrogen ions, which react with F− ions in the solution to form HF molecules. The pH will drop slightly if the buffer is within its capacity.

43. Write formula equations and net ionic equations for the hydrolysis of each salt in water.

a. sodium carbonate

$Na_2CO_3(s) + H_2O(l) \rightarrow NaHCO_3(aq) + NaOH(aq)$;
CO_3 2-(aq) + H $_2O(l) \rightarrow H_2CO_3$ −(aq) +$OH^-(aq)$





44. What acid-base indicators would be suitable for the neutralization reaction, whose titration curve is shown? Why?

Bromcresol purple or alizarin would be suitable because they would change color very near a pH 6.0 equivalence point.

45. How many milliliters of 0.225M HCl would be required to titrate 6.00 g of KOH?
475 HCl

46. Write the complete ionic and net ionic equations for the neutralization of H_2S by $Ca(OH)_2$.

a. Complete ionic equation:
$H_2S(aq) + Ca(OH)_2(aq) \rightarrow CaS(aq) + 2H_2O(l)$

b. Net ionic equation:
$H_2S\ (aq) + 2OH^-\ (aq) \rightarrow S2\text{-}\ (aq) + 2H_2O\ (l)$

47. Two students performed a titration to determine the concentration of a solution of acetic acid ($HC_2H_3O_2$). They used a known solution of 0.100M potassium hydroxide (KOH) as the standard solution.

a. Write a balanced formula equation for this acid-base reaction.

$HC_2H_3O_2(aq) + KOH(aq) \rightarrow KC_2H_3O_2(aq) + H_2O(l)$

b. From the balanced equation for this reaction, what is the mole ratio of the acid and the base?

1:1

c. Consider the strengths of the acid and base used in the titration. Would you expect the results of the reaction to be neutral, slightly acidic, or slightly basic? Explain.

The resulting salt is the product of a strong base and a weak acid. The results of its hydrolysis would be slightly basic.

d. Phenolphthalein was chosen as an indicator for the reaction. Why was phenolphthalein a good choice?

The end point for phenolphthalein occurs at a slightly basic pH. The neutralized solution will also be slightly basic.

e. Use the burette readings and mole ratio from the balanced equation, to fill out the remainder of the table and determine the concentration of acetic acid used.

Sample answers	
Initial Buret Volume (mL)	4.28
Final Buret Volume (mL)	34.30
Volume KOH (mL)	30.02
Volume $HC_2H_3O_2$ (mL)	25.00
Amount KOH (mol)	3.002 × 10-3
Amount $HC_2H_3O_2$ (mol)	3.002 × 10-3
Concentration $HC_2H_3O_2$ (M)	0.1201

48. The net ionic equation for the reaction between aqueous sulfuric acid and aqueous sodium hydroxide is ______ .

a. $H^+ (aq) + HSO_4^- (aq) + 2OH^- (aq) \rightarrow 2H_2O (l) + SO_4^{2-} (aq)$

b. $H^+ (aq) + HSO_4^- (aq) + 2Na^+ (aq) + 2OH^- (aq) \rightarrow 2H_2O (l) + 2Na^+ (aq) + SO_4^{2-} (aq)$

c. $2H^+ (aq) + SO_4^{2-} (aq) + 2Na^+ (aq) + 2OH^- (aq) \rightarrow 2H_2O (l) + 2Na^+ (aq) + SO_4^{2-} (aq)$

d. $SO_4^{2-} (aq) + 2Na^+ (aq) \rightarrow 2Na^+ (aq) + SO_4^{2-} (aq)$

e. $H^+ (aq) + OH^- (aq) \rightarrow H_2O(l)$

49. The net ionic equation for the reaction between aqueous nitric acid and aqueous sodium hydroxide is .

a. $HNO_3 (aq) + NaOH (aq) \rightarrow NaNO_3 (aq) + H_2O (l)$

b. $HNO_3 (aq) + OH^- (aq) \rightarrow NO_3^- (aq) + H_2O (l)$

c. $H^+ (aq) + HNO_3 (aq) + 2OH^- (aq) \rightarrow 2H_2O (l) + NO_3^- (aq)$

d. $H^+ (aq) + OH^- (aq) \rightarrow H_2O (l)$

e. $H^+ (aq) + Na^+ (aq) + OH^- (aq) \rightarrow H_2O (l) + Na^+ (aq)$